

COURSE TITLE : YARN FORMATION II
COURSE CODE : 4064
COURSE CATEGORY : B
PERIODS/WEEK : 4
PERIODS/SEMESTER : 56
CREDITS : 4

TIME SCHEDULE

Module	Topics	Periods
1	Draw frame , Lap Formers & Comber	14
2	Fly Frame	14
3	Ring Frame	14
4	Calculations	14
TOTAL		56

COURSE OUTCOME

Module	GO	Student will be able to
1	1	Know the need of drawing process and understand the working of Draw Frame
	2	Understand the Combing and comber preparatory process
2	1	Understand the working of Fly frame
	2	Know the winding, differential and building motion in Fly frame
3	1	Understand the working of Ring frame and drafting systems
	2	Know the Twisting, winding and building motion in ring frame
4	1	Understand the Draft and production of Draw frame.
	2	Know the production of Comber and combing preparatory machines
	3	Understand the Draft, Twist and production of Fly frame

SPECIFIC OUTCOME

MODULE I DRAW FRAME, LAP FORMERS & COMBER

- 1.1.0 **Know the need of drawing process and understand the working of Draw Frame**
- 1.1.1 Define drawing process and state the objectives
- 1.1.2 Define the principle of roller drafting
- 1.1.3 State the important terminologies in roller drafting
- 1.1.4 Explain the passage of material through a modern draw frame
- 1.1.5 State the functions of various parts of draw frame
- 1.1.6 Describe the working of modern drafting systems
- 1.1.7 State the importance of stop motion

- 1.1.8 Explain the working of electrical and electronic stop motions
- 1.1.9 List and explain the various top roller weighting systems.
- 1.1.10 State the functions of auto levelers
- 1.1.11 State the need for roller cleaning devices
- 1.1.12 List the defects in draw frame sliver - their causes and remedies

1.2.0 Understand the Combing and combing preparatory process

- 1.2.1 State the objectives of sliver lap machine
- 1.2.2 Explain the working of sliver lap machine
- 1.2.3 State the objectives of Ribbon lap machine
- 1.2.4 Explain the working of ribbon lap machine
- 1.2.5 Describe the working of Super lap machine
- 1.2.6 State the objectives of combing
- 1.2.7 Explain the working of modern comber
- 1.2.8 State the functions of important parts of comber
- 1.2.9 Illustrate the combing cycles
- 1.2.10 List combing defects – causes and remedies

MODULE II FLY FRAME

2.1.0 Understand the working of Fly frame

- 2.1.1 State the objectives of Fly frame
- 2.1.2 Explain the working of Fly frame
- 2.1.3 State the function of various parts of the Fly frame
- 2.1.4 Describe the drafting systems
- 2.1.5 Explain the various top roller weighting systems employed in fly frame
- 2.1.6 Describe the twisting operation in fly frame

2.2.0 Know the winding, differential and building motion in Fly frame

- 2.2.1 List the principles of winding
- 2.2.2 Differentiate Flyer leading and bobbin leading
- 2.2.3 Describe the principle of differential motion
- 2.2.4 Illustrate the working of T and S Differential motion
- 2.2.5 State the features of modern differential motion
- 2.2.6 State the functions of building motion in a fly frame
- 2.2.7 List the features of Rovematic speed frames
- 2.2.8 List various roving defects - their causes and remedies

MODULE III RING FRAME

3.1.0 Know the working of Ring frame and drafting systems

- 3.1.1 List the objectives of Ring frame
- 3.1.2 Describe the working Ring frame
- 3.1.3 State the functions of each part of Ring frame
- 3.1.4 Explain about the drafting systems SKF-PK-225 and L/R Pneumatic
- 3.1.5 State the functions of top and bottom roller clearers

3.2.0 Understand the Twisting, winding and building motion in ring frame

- 3.2.1 Explain the methods of twist insertion and winding in ring frame
- 3.2.2 Describe the method of producing 'Z' twist and 'S' twist yarn
- 3.2.3 Illustrate the different types of bobbin building
- 3.2.4 Explain working of building motion for weft wind
- 3.2.5 List the yarn defects - causes and remedies

MODULE IV CALCULATIONS

4.1.0 Know the Draft and production of Draw frame.

- 4.1.1 Determine the hank of sliver
- 4.1.2 Draw the gearing diagram of draw frame
- 4.1.3 Determine the draft and draft constant
- 4.1.4 Find the Production and efficiency of the draw frame
- 4.1.5 Estimate the can filling time in a draw frame

4.2.0 Understand the production of Comber and combing preparatory machines.

- 4.2.1 Determine the hank of sliver and hank of lap
- 4.2.2 Find the Production and efficiency of the sliver lap machine
- 4.2.3 Estimate the production and efficiency of comber

4.3.0 Know the Draft, Twist and production of Fly frame

- 4.3.1 Draw the gearing diagram of Fly frame
- 4.3.2 Determine the draft and draft constant
- 4.3.3 Estimate the twist and twist constant
- 4.3.4 Find the Production and efficiency of the Fly frame
- 4.3.5 Estimate the time required to build a full bobbin in a fly frame

CONTENT DETAILS

MODULE I DRAW FRAME , LAP FORMERS & COMBER

Define drawing process and state the objectives-Define the principle of roller drafting-State the important terminologies in roller drafting such as roller slip, floating fiber and drafting wave-Explain the passage of material through a modern draw frame-State the functions of various parts of draw frame-Describe the working of modern drafting systems-State the importance of stop motion-Explain the working of electrical and electronic stop motions-List and explain the various top roller weighting-State the functions of auto levelers-State the need for roller cleaning devices-List the defects in draw frame sliver - their causes and remedies-State the objectives of sliver lap machine-Explain the working of sliver lap machine-State the objectives of Ribbon lap machine-Explain the working of ribbon lap machine-Describe the working of Super lap machine-State the objectives of combing-Explain the working of modern comber-State the functions of important parts of comber-Illustrate the operations in combing cycle (feeding, nipping, combing, piecing and detaching)-List combing defects – causes and remedies.

MODULE II FLY FRAME

State the objectives of Fly frame-Explain the working of Fly frame-State the function of various parts of the Fly frame-Describe the drafting systems -Explain the various top roller weighting systems employed in fly frame-Describe the twisting operation in fly frame-List the principles of winding-Differentiate Flyer leading and bobbin leading-Describe the principle of differential motion-Illustrate the working of T and S Differential motion-State the features of modern differential motion-State the functions of building motion in a fly frame-List the features of Rovematic speed frame-List various roving defects - their causes and remedies.

MODULE III RING FRAME

List the objectives of Ring frame-Describe the working Ring frame- State the functions of each part of Ring frame such as creel, roving guide traverse, condenser, spacer, nose bar, apron cradle, lappet guide, ring, traveller, spindle, tin roller, traveller clearer, ring rail, spindle rail, pneumafil, separator, jockey pulley-Explain about the drafting systems SKF-PK-225 and L/R Pneumatic-State the functions of top and bottom roller clearers-Explain the methods of twist insertion and winding in ring frame-Describe the method of producing 'Z' twist and 'S' twist yarn-Illustrate the different types of bobbin building-Explain working of building motion for weft wind-List the yarn defects - causes and remedies.

MODULE IV CALCULATIONS

Determine the hank of sliver-Draw the gearing diagram of draw frame-Determine the draft and draft constant-Find the Production and efficiency of the draw frame-Estimate the can filling time in a draw frame-Determine the hank of sliver and hank of lap-Find the Production and efficiency of the sliver lap machine-Estimate the production and efficiency of comber, Calculate waste percentage, determine the hank of sliver -Draw the gearing diagram of Fly frame-Determine the draft and draft constant-Estimate the twist and twist constant-Find the Production and efficiency of the Fly frame-Estimate the time required to build a full bobbin in a fly frame.

REFERENCE BOOKS

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| 1. Taggart | Cotton Spinning | - Mahajan Publishers |
| 2. Textile institute | Manual of Cotton spinning | |
| 3. Merrill | Cotton Drafting, Roving, Combing Cotton spinning | |
| 4. Pattabiraman | Essential elements of cotton spinning | |
| 5. Taggart | Cotton Spinning Calculations | |